

Asymptotic expansions and inversion for A124380

Research report

1 October 2026

Abstract

The coefficients of $\sum_{k \geq 0} z^k \prod_{j=1}^k (1 + jz)$ admit an exact decomposition into a positive moment and a signed moment. This identity yields a multiplicative asymptotic equivalent, an arbitrary fixed-order relative expansion, and a separately controlled oscillatory expansion for the exponentially smaller signed term. Both sectors have explicit finite coefficient recurrences. We also invert the positive moment through a Lambert W normalization, give five explicit inverse coefficients and an arbitrary-order recurrence, and explain the additional certification needed for integer thresholds. The proof uses real Laplace estimates and Gaussian moments; no convergence of the formal series, complete exponential classification, or publication-priority claim is made.

Provenance, status and reading conventions

[write] **Added 2 October 2026, batch 77.** *Provenance.* This report was built from one manuscript: batch 77, manuscript 06 of the repository’s incoming-reports channel, archive `a124380-reproducibility.zip` (wrapper directory `a124380-asymptotics/`, main file `a124380-asymptotics.tex`, 593 lines, with a fourteen-page PDF that is not shipped), arrival commit `096ee7b87`, placed in `4f11bc9c0`. From the abstract to the bibliography the text is the delivered manuscript, unchanged except for the label prefix `sma:` and the notes marked [write]; section, theorem and equation numbers are the manuscript’s own. *Pin.* The delivery names no ProveIt commit and continues no repository path; at placement no repository report treated A124380. *Status.* An unrefereed research report; the delivery names no author or tool (its author line reads “Research report”), and its “mathematical review” record (shipped as `data/mathematical-review.json`) names no reviewer. Nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status. On 2 October 2026, on a scratch copy, all four delivered checkers (`independent_checks.py`, `verify_all.py`, `verify_oscillatory.py`, `check_sector.py`) passed, and they and the three coefficient generators reproduced all eight recorded outputs identically apart from line endings. An independent computation of a_n from (1.1) reproduced the fifteen initial terms below, and at $n = 3000$ the residual $x(a_n/M_n - 1) = 4.627$ against $q_1(L) = 4.341$, with $x^2(a_n/M_n - 1 - q_1/x) = 11.06$ against $q_2(L) = 10.59$, consistent with (5.4) and a next term of size $O((1 + L)^9/x)$. These replays check the transcription and the implementation, not the proofs.

[write] **Reading conventions.** The manuscript reuses several letters. No symbol of the delivered text was renamed; the table fixes the readings and the tempting false ones.

symbol	meanings, by section
x, L, Λ	$x = \sqrt{n/2}$ and $L = \log x$ throughout the forward expansions; on the inverse side (Section 7) X and $\Lambda = \log X$ are the Lambert variables of (7.3), and Appendix A writes L for Λ .
y	the local offset $t = x + y$ in Sections 5–6; in Section 7 the <i>logarithmic</i> level $\log I_{\nu(y)} = y$, so $T(y)$ compares a_n with e^y , not with y .
B, B_{2r}	$B = x^2(2L - 1)$; B_{2r} are Bernoulli numbers.
$\mathcal{M}_n, M_n, M$	$\mathcal{M}_n$ is the leading factor (1.3) and $\mathcal{M}_n^-$ its oscillatory analogue; $M_n(a)$ is the auxiliary entire function (2.5); $M = \sum m_j$ in (4.1), but M is a truncation order in Theorem 7.1.
m	a requested weight in the proof of Theorem 4.1; m_d are normal moments (5.3); in Section 7.3 m is a certified lower bound for $(\log I_s)'$.
μ	$\mu = (L + 2)/4$ in (5.3); μ_m in (6.1) are complex Fourier–Gaussian moments.
c, C	$c = \frac{1}{2} - \log 2$ in Section 7 and Appendix A; elsewhere c, C, C_K are unspecified positive constants. C_r in (4.1) are saddle coefficients.
A	$A_n(u)$ is the polynomial (2.1); $A(s)$ the phase remainder in the proof of Theorem 4.1; $A_{n,k}$ the triangle (8.1).
R	$R(z; r)$ in the proof of Theorem 2.1; R a truncation order; $R_j^\pm$ the phase polynomials (5.1), (6.5).
K	a radius $ a \leq K$ in the proof of Theorem 2.1; $K(z, v; \Lambda)$ the formal expression (7.4).
q_j, P_j	$q_j = q_j^+$ (written without the sign in Theorem 5.1 and (6.7)) and q_j^- ; P_j the logarithmic coefficients (5.5).
$\mathbb{E}, \mathcal{E}$	$\mathbb{E}$ is a normal expectation; $\mathcal{E}$ is a linear functional with a complex mean, not a probability measure.
N	$N_R(y)$ an implicit inverse (7.9); N_2, N_3, N_4 the numerator polynomials of Appendix A.
ν, T	$\nu(y)$ inverts the continuation $s \mapsto \log I_s$, not an interpolation of a_n ; $T(y)$ is the integer threshold.

1 Sequence and main conclusions

Define $a_0 = 1$ and, as a formal power series,

$$\sum_{n \geq 0} a_n z^n = \sum_{k \geq 0} z^k \prod_{j=1}^k (1 + jz). \quad (1.1)$$

These are OEIS A124380, beginning

$$1, 1, 2, 4, 9, 22, 57, 157, 453, 1368, 4296, 13995, 47138, 163779, 585741.$$

The series is used formally; none of the arguments requires a positive radius of convergence. For $n \rightarrow \infty$ put

$$x = \sqrt{n/2}, \quad L = \log x, \quad B = x^2(2L - 1). \quad (1.2)$$

The principal equivalent proved below is

$$a_n \sim \mathcal{M}_n, \quad \mathcal{M}_n = \frac{e^{1/2}}{2} x \exp \left\{ B + x(L + 1) + \frac{L^2}{8} \right\}. \quad (1.3)$$

In particular,

$$\log a_n = B + x(L + 1) + \frac{L^2}{8} + L + \frac{1}{2} - \log 2 + O\left(\frac{(1 + L)^3}{x}\right). \quad (1.4)$$

This proves substantially more than the logarithmic conjecture recorded in the OEIS entry [1]: the factor in (1.3) has relative error tending to zero. The terms of size $\sqrt{n} \log n$, $(\log n)^2$, and the multiplicative prefactor matter for this conclusion.

There is also an exact identity

$$a_n = I_n + (-1)^n J_n, \quad (1.5)$$

where $I_n > 0$ and J_n is signed. The signed contribution satisfies

$$\frac{|J_n|}{I_n} \leq \exp\{-2x(L+1) + O(L^2)\}, \quad (1.6)$$

$$J_n = \mathcal{M}_n^- \left\{ \sin \theta_n + O\left(\frac{(1+L)^3}{x}\right) \right\}, \quad (1.7)$$

$$\mathcal{M}_n^- = e^{1/2 - \pi^2/8} x \exp\left\{B - x(L+1) + \frac{L^2}{8}\right\}, \quad \theta_n = \pi x - \frac{\pi(L+2)}{4}.$$

The error in (1.7) is measured against $\mathcal{M}_n^-$, not against J_n ; it remains valid near zeros of the sine. Both (1.3) and (1.7) extend to every fixed algebraic order. A finite algebraic approximation to I_n generally has an error much larger than J_n . The exact subtraction $a_n - I_n = (-1)^n J_n$ is therefore essential when interpreting the smaller sector.

1.1 Established combinatorial models and asymptotic background

Chen, Fan, and Zhao [2], Theorem 1.1, already identify (1.1) with partial matchings avoiding neighbor alignments. Their Theorem 2.1 gives the corresponding triangular recurrence. This natural model dates to their 2010 preprint. Their displayed initial list has 4290 where their generating function gives 4296; the identification here uses the generating function.

Cerbai, Claesson, and Sagan [3], Theorem 4.5 and Lemma 4.6, give a later model consisting of restricted-growth words

$$1B_1 2B_2 \cdots kB_k,$$

where B_i is empty or a single letter in $\{1, \dots, i\}$. They also identify the stabilized large-difference limit of self-modified difference ascent sequences. These models and their historical precedence are established work, not discoveries claimed by this report.

Temme [4] gives important uniform asymptotic estimates for Stirling numbers; the first-kind analysis includes gamma-ratio saddles. The method here first sums the relevant diagonal exactly into real moments, then applies ordinary Laplace expansion. A targeted literature check did not locate a sequence-specific relative asymptotic theorem of the form proved here. That observation does not establish publication priority or exhaust the literature.

2 An exact signed moment decomposition

Let e_j denote the elementary symmetric polynomial, and let square brackets with two entries denote unsigned Stirling numbers of the first kind. Introduce

$$A_n(u) = \sum_{k=\lceil n/2 \rceil}^n e_{n-k}(1, \dots, k) u^k, \quad a_n = A_n(1) = \sum_k \left[\begin{matrix} k+1 \\ 2k-n+1 \end{matrix} \right]. \quad (2.1)$$

All sums in this section are finite when indexed by a fixed n .

Theorem 2.1. *For every integer $n \geq 0$,*

$$a_n = I_n + (-1)^n J_n, \quad (2.2)$$

$$I_n = \int_0^\infty \frac{t^{n+2t+1} e^{-t^2}}{\Gamma(1+t)} dt, \quad (2.3)$$

$$J_n = \int_0^\infty t^{n-2t+1} e^{-t^2} \frac{1}{\Gamma(1-t)} dt. \quad (2.4)$$

Both integrals are absolutely convergent. The reciprocal gamma function in (2.4) is entire and vanishes at positive integral t ; those points are not singularities of the integrand.

Proof. For $r > 0$ use its real logarithm and set

$$R(z; r) = \frac{r^z}{\Gamma(1+z)}.$$

For $a \in \mathbb{C}$, define

$$M_n(a) = \frac{a^n}{2} \int_0^\infty e^{-r} r^{n/2} \{R(a\sqrt{r}; r) + (-1)^n R(-a\sqrt{r}; r)\} dr. \quad (2.5)$$

This is entire in a . Indeed, on $|a| \leq K$, the standard global reciprocal-gamma bound gives, for $r \geq 1$, an integrable majorant of the form

$$C_K r^{n/2} \exp\{-r + C_K \sqrt{r} \log(2+r)\}.$$

For $0 < r \leq 1$, the arguments of reciprocal gamma lie in a compact disc, while $\sqrt{r} |\log r| \rightarrow 0$. Thus the remaining factors are uniformly bounded. Dominated holomorphic parameter integration and Cauchy coefficient extraction apply.

Only Taylor indices $m \geq 0$ with $n+m$ even survive the parity sum. Put $k = (n+m)/2$. The coefficient of $a^{n+m} = a^{2k}$ in (2.5) is

$$\begin{aligned} \int_0^\infty e^{-r} r^k [z^m] \frac{r^z}{\Gamma(1+z)} dr &= [z^m] \frac{\Gamma(k+1+z)}{\Gamma(1+z)} \\ &= [z^m] \prod_{j=1}^k (j+z) = e_{n-k}(1, \dots, k). \end{aligned}$$

The gamma-integral exchange is justified uniformly on, for example, $|z| < 1/2$. The product has degree k , so the coefficient is zero if $m > k$, equivalently $k > n$. Consequently the entire function $M_n(a)$ equals the finite polynomial $A_n(a^2)$. Set $a = 1$ and substitute $r = t^2$ in (2.5); the factor $2t$ cancels its $1/2$ and yields (2.2). $\square$

For completeness, the same identity gives, for real $u > 0$,

$$A_n(u) = \int_0^\infty e^{-t^2/u} \left\{ \frac{u^{-1-t} t^{n+2t+1}}{\Gamma(1+t)} + (-1)^n \frac{u^{-1+t} t^{n-2t+1}}{\Gamma(1-t)} \right\} dt. \quad (2.6)$$

No uniform moving- u asymptotic assertion is needed in this report.

3 Saddles and the exponentially small scale

Define the two real phases

$$\Phi_n(t) = (n + 2t + 1) \log t - t^2 - \log \Gamma(1 + t), \quad (3.1)$$

$$\Psi_n(t) = (n - 2t + 1) \log t - t^2 + \log \Gamma(t). \quad (3.2)$$

Reflection gives the exact expression

$$J_n = \frac{1}{\pi} \int_0^\infty e^{\Psi_n(t)} \sin(\pi t) dt. \quad (3.3)$$

Here ψ and ψ_1 below denote the digamma and trigamma functions, respectively.

Lemma 3.1. *For $n \geq 1$, each phase has a unique positive saddle, denoted by τ for Φ_n and σ for Ψ_n . They satisfy*

$$\Phi_n''(t) \leq -2 + \frac{1}{n+1} < -1, \quad (3.4)$$

$$\Psi_n''(t) < -2, \quad (3.5)$$

$$\tau = x + \frac{L+2}{4} + O(L^2/x), \quad \sigma = x - \frac{L+2}{4} + O(L^2/x). \quad (3.6)$$

At either saddle, the negative second derivative tends to 4. For every fixed $j \geq 3$, the j th derivative is $O_j(x^{2-j})$, uniformly on the interval from one half to three halves of the corresponding saddle.

Proof. Direct differentiation gives

$$\Phi_n''(t) = -\frac{n+1}{t^2} + \frac{2}{t} - 2 - \psi_1(1+t).$$

The quadratic $-(n+1)v^2 + 2v$ is at most $1/(n+1)$, proving (3.4). For the other phase, the series for trigamma and an integral comparison give

$$\psi_1(t) = \frac{1}{t^2} + \sum_{k \geq 1} \frac{1}{(t+k)^2} < \frac{1}{t^2} + \int_0^\infty \frac{ds}{(t+s)^2} = \frac{1}{t^2} + \frac{1}{t}.$$

Hence

$$\Psi_n''(t) = -\frac{n+1}{t^2} - \frac{2}{t} - 2 + \psi_1(t) < -\frac{n}{t^2} - \frac{1}{t} - 2.$$

Both first derivatives tend to $+\infty$ at zero and to $-\infty$ at infinity. Strict concavity yields existence and uniqueness. The exact positive saddle equation is

$$n = 2\tau^2 - 2\tau \log \tau - 2\tau + \tau\psi(1+\tau) - 1. \quad (3.7)$$

Stirling's formula and its differentiated forms first locate both saddles at $x + O(L)$ and then give (3.6), the curvature limit, and the derivative estimates. $\square$

Proposition 3.2. *As $n \rightarrow \infty$, (1.6) holds. In particular J_n/I_n is smaller than every fixed inverse power of n .*

Proof. For $t \geq 1$, Stirling's formula gives

$$\begin{aligned}\Phi_n(t) &= (n + t + \tfrac{1}{2}) \log t - t^2 + t - \tfrac{1}{2} \log(2\pi) + O(1/t), \\ \Psi_n(t) &= (n - t + \tfrac{1}{2}) \log t - t^2 - t + \tfrac{1}{2} \log(2\pi) + O(1/t).\end{aligned}$$

Evaluation at x , together with the $O(L)$ saddle displacements and bounded saddle curvatures, gives

$$\Phi_n(\tau) = B + x(L + 1) + O(L^2), \quad \Psi_n(\sigma) = B - x(L + 1) + O(L^2).$$

A fixed-width interval about τ gives $I_n \geq ce^{\Phi_n(\tau)}$ for all sufficiently large n . Global concavity gives

$$|J_n| \leq \pi^{-1} \int_0^\infty e^{\Psi_n(t)} dt \leq \pi^{-1/2} e^{\Psi_n(\sigma)}.$$

Their ratio proves the claim. This is an upper bound and does not exclude further oscillatory cancellation. $\square$

4 Full expansion at the positive saddle

Set

$$H_+ = -\Phi_n''(\tau), \quad \lambda_j^+ = \Phi_n^{(j)}(\tau) \quad (j \geq 3).$$

For $r \geq 0$ define the finite expression

$$C_r = \sum_{\substack{m_3, \dots, m_{2r+2} \geq 0 \\ \sum_{j=3}^{2r+2} (j-2)m_j = 2r}} \frac{(2r + 2M - 1)!!}{H_+^{r+M}} \prod_{j=3}^{2r+2} \frac{(\lambda_j^+ / j!)^{m_j}}{m_j!}, \quad M = \sum_j m_j, \quad (4.1)$$

with $C_0 = 1$ and $(-1)!! = 1$.

Theorem 4.1. *For each fixed $R \geq 0$, $C_r = O_r(\tau^{-2r})$ and*

$$I_n = e^{\Phi_n(\tau)} \sqrt{\frac{2\pi}{H_+}} \left\{ \sum_{r=0}^R C_r + O_R(\tau^{-2R-2}) \right\}. \quad (4.2)$$

The same formula holds with a_n in place of I_n . Its constants may depend on R but not on n .

For example, writing $\lambda_j = \lambda_j^+$ and $H = H_+$,

$$C_1 = \frac{\lambda_4}{8H^2} + \frac{5\lambda_3^2}{24H^3}, \quad (4.3)$$

$$C_2 = \frac{\lambda_6}{48H^3} + \frac{7\lambda_3\lambda_5}{48H^4} + \frac{35\lambda_4^2}{384H^4} + \frac{35\lambda_3^2\lambda_4}{64H^5} + \frac{385\lambda_3^4}{1152H^6}. \quad (4.4)$$

Proof. Put $t = \tau + s$. By global concavity,

$$\Phi_n(\tau + s) - \Phi_n(\tau) \leq -\frac{s^2}{2}.$$

Fix a requested weight m . On $|s| \leq \delta\tau$, with sufficiently small fixed $\delta > 0$, write

$$A(s) = \Phi_n(\tau + s) - \Phi_n(\tau) + H_+ s^2 / 2.$$

The third derivative bound gives $|A(s)| \leq C|s|^3/\tau \leq C\delta s^2$. Choose δ so that this is at most $H_+ s^2/4$. Taylor expansion of the phase through derivative $m+2$ has remainder $O_m(\tau^{-m-1}|s|^{m+3})$. Expanding the residual exponential through power m and using its integral remainder bounds the additional error by τ^{-m-1} times a fixed polynomial in $|s|$ times $e^{H_+ s^2/4}$. This is integrable against $e^{-H_+ s^2/2}$. A factor $\lambda_j s^j/j!$ has weight $j-2$, so the finitely many retained monomials of weight greater than m also have integrated size $O_m(\tau^{-m-1})$.

The complementary tails $|s| > \delta\tau$ are $O(e^{-c\tau^2})$ after division by $e^{\Phi_n(\tau)}$, by global concavity. Extending the retained polynomial Gaussian integrals to the whole real line has the same negligible error. Take $m = 2R + 1$. A monomial indexed by (m_j) has power $\sum j m_j = \sum (j-2)m_j + 2M$ in s . Odd weights therefore integrate to zero. The normal moment for weight $2r$ is $(2r + 2M - 1)!!/H_+^{r+M}$, which gives (4.1). Proposition 3.2 absorbs J_n when replacing I_n by a_n . $\square$

5 Explicit powers and polynomial logarithms

The positive saddle form has log-free remainder estimates. An alternative makes every coefficient elementary in L . For $j \geq 1$ define

$$R_j^+(y) = (-1)^{j+1} \left\{ \frac{2y^{j+2}}{j+2} + \frac{y^{j+1}}{j(j+1)} + \frac{y^j}{2j} - \sum_{r=1}^{\lfloor (j+1)/2 \rfloor} \frac{B_{2r}}{2r(2r-1)} \binom{j-1}{2r-2} y^{j-2r+1} \right\}, \quad (5.1)$$

where B_{2r} are Bernoulli numbers. Starting with $Q_0^+ = 1$, put

$$Q_j^+(y) = \frac{1}{j} \sum_{\ell=1}^j \ell R_\ell^+(y) Q_{j-\ell}^+(y), \quad q_j(L) = \mathbb{E} Q_j^+(Y), \quad (5.2)$$

where Y is normal with mean $(L+2)/4$ and variance $1/4$. Its moments are obtained without integration from

$$m_0 = 1, \quad m_1 = \mu, \quad m_d = \mu m_{d-1} + \frac{d-1}{4} m_{d-2}, \quad \mu = \frac{L+2}{4}. \quad (5.3)$$

Theorem 5.1. *For every fixed $R \geq 0$, $q_0 = 1$, $\deg q_j \leq 3j$, and*

$$I_n = \mathcal{M}_n \left\{ \sum_{j=0}^R \frac{q_j(L)}{x^j} + O_R \left(\frac{(1+L)^{3R+3}}{x^{R+1}} \right) \right\}. \quad (5.4)$$

The same expansion holds for a_n . Define $P_0 = 0$ and

$$P_j = q_j - \frac{1}{j} \sum_{\ell=1}^{j-1} \ell P_\ell q_{j-\ell}. \quad (5.5)$$

Then

$$\log I_n = \log \mathcal{M}_n + \sum_{j=1}^R \frac{P_j(L)}{x^j} + O_R \left(\frac{(1+L)^{3R+3}}{x^{R+1}} \right), \quad (5.6)$$

and likewise for $\log a_n$.

Proof. Set $t = x + y$ in (2.3). Stirling's expansion for $\log \Gamma(1 + t)$, with its fixed-order real remainder, gives the residual phase

$$-2y^2 + (L + 2)y + \sum_{j \geq 1} R_j^+(y)x^{-j}, \quad (5.7)$$

after factoring $\exp\{B + x(L + 1) + L/2\}/\sqrt{2\pi}$. The sum here is formal: at any fixed order it is replaced by a finite Taylor expansion with a remainder. Expanding $\log(1 + y/x)$ and the reciprocal powers in the Stirling correction gives exactly (5.1). Completing the square in the order-zero phase produces mean $(L + 2)/4$, variance $1/4$, and the factor $\mathcal{M}_n$. Exponentiating the finite residual series gives (5.2).

Each R_j^+ has degree $j + 2$, and each Q_j^+ has degree at most $3j$. For the error, put $z = y - \mu$, where $\mu = (L + 2)/4$, and work on $|z| \leq \delta x$ with small fixed δ . The phase remainder after R corrections is

$$O_R\left(x^{-R-1}(1 + |y|^{R+3})\right).$$

The full residual after the quadratic part has absolute value at most $C(1 + |y|^3)/x$. Since $|y|^3 \leq 4(|\mu|^3 + |z|^3)$, its exponential is bounded by a constant times $e^{C'\delta z^2}$; the other factor $e^{O((1+|\mu|^3)/x)}$ is bounded. Choosing δ small absorbs this into the centered Gaussian. The exponential Taylor remainder is therefore bounded, after integration, by x^{-R-1} times Gaussian moments of degree at most $3R + 3$. A discarded monomial of weight $w > R$ has polynomial degree at most $3w$; its integrated size is $O_R((1 + L)^{3w}x^{-w})$. There are only finitely many such monomials at the chosen expansion depth, and their excess weights are absorbed using $(1 + L)^3/x \rightarrow 0$. This leaves the common bound $O_R((1 + L)^{3R+3}x^{-R-1})$. The exact saddle differs from $x + \mu$ by $O(L^2/x)$, so global concavity makes the omitted tails exponentially small. This proves (5.4). Formal logarithm extraction yields (5.5); finite-order Taylor control of $\log(1 + v)$ preserves the stated conservative remainder. The signed term is negligible by Proposition 3.2. $\square$

The first relative coefficients are

$$q_1 = \frac{L^3 + 9L^2 + 48L + 72}{96}, \quad (5.8)$$

$$q_2 = \frac{L^6 + 18L^5 + 177L^4 + 1104L^3 + 4032L^2 + 9024L + 8544}{18432}. \quad (5.9)$$

The logarithmic coefficients are considerably shorter:

$$\begin{aligned} P_1 &= \frac{L^3 + 9L^2 + 48L + 72}{96}, \\ P_2 &= \frac{2L^3 + 9L^2 + 44L + 70}{384}, \\ P_3 &= -\frac{9L^5 + 105L^4 + 540L^3 + 2580L^2 + 6000L + 5072}{92160}, \\ P_4 &= -\frac{12L^5 + 90L^4 + 420L^3 + 1215L^2 + 2652L + 2323}{92160}. \end{aligned} \quad (5.10)$$

These are finite-order asymptotic coefficients. The report makes no convergence claim for the infinite formal series.

Corollary 5.2. *The sequence is eventually strictly log-convex, and*

$$\log \frac{a_{n-1}a_{n+1}}{a_n^2} \sim \frac{1}{2n}.$$

Proof. Use the logarithmic expansion through P_2 and denote its explicit smooth part by $F(n)$. Its second derivative is $F''(n) = 1/(2n) + O(\log n/n^{3/2})$. Its centered second difference is the integral of F'' against the triangular kernel on $[-1, 1]$, so it is asymptotic to $1/(2n)$. The three-point difference of the remainder is $O((\log n)^9/n^{3/2}) = o(1/n)$ simply by bounding its three values. No differentiation of an unknown remainder is used. $\square$

6 The oscillatory sector to every fixed order

6.1 Expansion at its real saddle

Let

$$H_- = -\Psi_n''(\sigma), \quad \lambda_j^- = \Psi_n^{(j)}(\sigma), \quad j \geq 3.$$

Define complex Gaussian moments by

$$\mu_0 = 1, \quad \mu_1 = \frac{i\pi}{H_-}, \quad \mu_m = \frac{i\pi}{H_-} \mu_{m-1} + \frac{m-1}{H_-} \mu_{m-2}. \quad (6.1)$$

For $d \geq 0$, put

$$D_d = \sum_{\substack{m_3, \dots, m_{d+2} \geq 0 \\ \sum (j-2)m_j = d}} \mu_{\sum j m_j} \prod_{j=3}^{d+2} \frac{(\lambda_j^-/j!)^{m_j}}{m_j!}, \quad D_0 = 1. \quad (6.2)$$

Theorem 6.1. *For every fixed $R \geq 0$, $D_d = O_d(\sigma^{-d})$ and*

$$J_n = \frac{e^{\Psi_n(\sigma)}}{\pi} \sqrt{\frac{2\pi}{H_-}} e^{-\pi^2/(2H_-)} \operatorname{Im} \left\{ e^{i\pi\sigma} \sum_{d=0}^R D_d \right\} + O_R(e^{\Psi_n(\sigma)} \sigma^{-R-1}). \quad (6.3)$$

This is an absolute, amplitude-scaled error estimate, including near cancellations of the oscillation.

Proof. Use (3.3), shift $t = \sigma + s$, and replace the sine by the imaginary part of $e^{i\pi(\sigma+s)}$. The Gaussian Fourier integral is

$$\int_{\mathbb{R}} e^{-H_- s^2/2 + i\pi s} ds = \sqrt{\frac{2\pi}{H_-}} e^{-\pi^2/(2H_-)}.$$

After division by this factor, its polynomial moments are exactly (6.1), by differentiation or integration by parts. The Taylor and exponential expansion used in Theorem 4.1 applies unchanged in absolute value, because $|e^{i\pi s}| = 1$. Global concavity of Ψ_n bounds the complementary tails. Keeping weights at most R leaves $O_R(\sigma^{-R-1})$ in the normalized integral. Unlike the positive sector, odd powers need not disappear: their Fourier moments are imaginary. This gives (6.2) and (6.3). $\square$

In particular, writing $\lambda_3 = \lambda_3^-$ and $H = H_-$,

$$J_n = \frac{e^{\Psi_n(\sigma)}}{\pi} \sqrt{\frac{2\pi}{H}} e^{-\pi^2/(2H)} \left\{ \sin(\pi\sigma) + \lambda_3 \left(\frac{\pi}{2H^2} - \frac{\pi^3}{6H^3} \right) \cos(\pi\sigma) + O(\sigma^{-2}) \right\}. \quad (6.4)$$

Since $\lambda_3 = 4/\sigma + O(\log \sigma/\sigma^2)$, the cosine coefficient is

$$\frac{\pi(12 - \pi^2)}{96\sigma} + O\left(\frac{\log \sigma}{\sigma^2}\right).$$

6.2 Elementary coefficient recurrence for the oscillation

For $j \geq 1$ define

$$R_j^-(y) = (-1)^{j+1} \left\{ \frac{2y^{j+2}}{j+2} - \frac{y^{j+1}}{j(j+1)} + \frac{y^j}{2j} + \sum_{r=1}^{\lfloor (j+1)/2 \rfloor} \frac{B_{2r}}{2r(2r-1)} \binom{j-1}{2r-2} y^{j-2r+1} \right\}. \quad (6.5)$$

Set $Q_0^- = 1$ and use the exponential recurrence

$$Q_j^-(y) = \frac{1}{j} \sum_{\ell=1}^j \ell R_\ell^-(y) Q_{j-\ell}^-(y).$$

For the complex mean

$$\zeta = -\frac{L+2}{4} + \frac{i\pi}{4},$$

define a linear moment functional by $\mathcal{E}(1) = 1$, $\mathcal{E}(y) = \zeta$, and

$$\mathcal{E}(y^m) = \zeta \mathcal{E}(y^{m-1}) + \frac{m-1}{4} \mathcal{E}(y^{m-2}).$$

This is a polynomial functional, not a probability measure with complex outcomes. Put $q_j^-(L) = \mathcal{E}(Q_j^-(y))$.

Theorem 6.2. *For every fixed $R \geq 0$, $q_0^- = 1$, $\deg_L q_j^- \leq 3j$, and*

$$J_n = \mathcal{M}_n^- \left\{ \operatorname{Im} \left(e^{i\theta_n} \sum_{j=0}^R \frac{q_j^-(L)}{x^j} \right) + O_R \left(\frac{(1+L)^{3R+3}}{x^{R+1}} \right) \right\}. \quad (6.6)$$

The constants may depend on R ; π is fixed.

Proof. In (3.3), put $t = x + y$. The Stirling form for Ψ_n gives the order-zero residual phase $-2y^2 - (L+2)y$, with prefactor

$$\frac{\sqrt{2\pi}}{\pi} \exp\{B - x(L+1) + L/2\}.$$

The higher residual phases are precisely (6.5): the $t \log t$ and Bernoulli terms reverse signs relative to the positive sector. Completing the real square gives center $-(L+2)/4$. Taking the Gaussian Fourier transform supplies $e^{-\pi^2/8}$ and tilts the polynomial moments to the complex mean ζ . The resulting prefactor is $\mathcal{M}_n^-$ and the phase is θ_n .

The finite Taylor and exponential remainder proof of Theorem 5.1 applies to absolute values, with the same polynomial degree bound and Gaussian moments. The Fourier normalization is a fixed nonzero constant. This proves the stated amplitude-scaled estimate. $\square$

There is a useful exact identity between the two finite coefficient families:

$$q_j^-(L) = (-1)^j q_j^+(L - i\pi). \quad (6.7)$$

Indeed, $R_j^-(-y) = (-1)^j R_j^+(y)$, and the exponential recurrence gives $Q_j^-(-y) = (-1)^j Q_j^+(y)$. Reflecting the complex moment mean gives $(L - i\pi + 2)/4$, proving (6.7) algebraically. No integration contour is moved.

Writing $q_1^- = U_1 + iV_1$, the first elementary correction is

$$\begin{aligned} U_1(L) &= \frac{-L^3 - 9L^2 + (3\pi^2 - 48)L + 9\pi^2 - 72}{96}, \\ V_1(L) &= \frac{\pi(3L^2 + 18L + 48 - \pi^2)}{96}, \\ J_n &= \mathcal{M}_n^- \left\{ \sin \theta_n + \frac{U_1 \sin \theta_n + V_1 \cos \theta_n}{x} + O\left(\frac{(1+L)^6}{x^2}\right) \right\}. \end{aligned} \quad (6.8)$$

In particular,

$$\frac{J_n}{\mathcal{M}_n} = 2e^{-\pi^2/8 - 2x(L+1)} \left\{ \sin \theta_n + O\left(\frac{(1+L)^3}{x}\right) \right\}. \quad (6.9)$$

Equations (2.2), (5.4), and (6.6) give two separately controlled asymptotic sectors under an exact decomposition. They do not assert a relative equivalent for J_n uniformly in n , convergence, optimal truncation, or a complete resurgent or exponential-transseries classification.

7 A canonical smooth inverse

For real $s > -2$ define I_s by (2.3) with n replaced by s . The condition $s > -2$ ensures integrability at zero. All fixed-order positive-sector expansions above hold as $s \rightarrow \infty$ through real values. Differentiation under the integral and saddle concentration yield

$$\frac{d}{ds} \log I_s = \log \sqrt{s/2} + O\left(\frac{\log s}{\sqrt{s}}\right) > 0 \quad (7.1)$$

for all sufficiently large s . For example, this follows by inserting $\log t$ into the normalized Laplace integral; the mass is concentrated within bounded distance of τ_s . Thus, for all sufficiently large y , there is a unique large solution $\nu(y)$ to

$$\log I_{\nu(y)} = y. \quad (7.2)$$

This specifies the continuation being inverted; it does not assume an arbitrary interpolation of a_n .

7.1 Lambert normalization and a finite recurrence

Let W denote the positive real Lambert function and put

$$X = \sqrt{\frac{y}{W(y/e)}}, \quad \Lambda = \log X, \quad c = \frac{1}{2} - \log 2. \quad (7.3)$$

Then $X^2(2\Lambda - 1) = y$ exactly. Define the following formal expression in z and v :

$$\begin{aligned} K(z, v; \Lambda) &= (1+v)^2 \{2\Lambda + 2\log(1+v) - 1\} - (2\Lambda - 1) \\ &\quad + z(1+v) \{\Lambda + \log(1+v) + 1\} \\ &\quad + z^2 \left\{ \frac{(\Lambda + \log(1+v))^2}{8} + \Lambda + \log(1+v) + c \right\} \\ &\quad + \sum_{j \geq 1} z^{j+2} (1+v)^{-j} P_j(\Lambda + \log(1+v)). \end{aligned} \quad (7.4)$$

Every requested coefficient uses only finitely many P_j . Recursively define

$$d_k(\Lambda) = -\frac{1}{4\Lambda}[z^{k+1}]K\left(z, \sum_{j=0}^{k-1} d_j(\Lambda)z^{j+1}; \Lambda\right), \quad k \geq 0, \quad (7.5)$$

where the empty sum is zero. Finally put

$$b_k = 4d_k + 2 \sum_{i=0}^{k-1} d_i d_{k-1-i}. \quad (7.6)$$

Thus all coefficients are rational functions of Λ and c and are computable by exact arithmetic.

Theorem 7.1. *For every fixed integer $M \geq 2$,*

$$\nu(y) = 2X^2 + \sum_{j=0}^M b_j(\Lambda)X^{1-j} + O_M\left(\frac{(1+\Lambda)^{3M-1}}{X^M}\right). \quad (7.7)$$

The first two levels, with their sharper simple remainder, are

$$\begin{aligned} \nu(y) = & 2X^2 - X(1 + 1/\Lambda) - \Lambda/8 - 3/4 \\ & + \frac{\log 2 + 1/8}{\Lambda} + \frac{1}{4\Lambda^2} - \frac{1}{8\Lambda^3} + O(\Lambda^2/X). \end{aligned} \quad (7.8)$$

The three further explicit terms b_2/X , b_3/X^2 , and b_4/X^3 are given in Appendix A.

Proof. Write $x = X(1 + v)$ and multiply the forward logarithmic expansion minus y by X^{-2} . With $z = X^{-1}$, the resulting formal expression is exactly (7.4). Its linear coefficient in v at $z = v = 0$ is 4Λ . At degree z^{k+1} , the previously undetermined coefficient d_k therefore occurs only as $4\Lambda d_k$. This proves the triangular cancellation rule (7.5). Squaring $x = X + \sum d_j X^{-j}$ and multiplying by 2 gives (7.6).

For a remainder proof, truncate the forward expansion after P_{M-1} and use Theorem 5.1, giving log-error $O_M((1+\Lambda)^{3M}X^{-M})$. The recurrence has $d_0 = O(1)$, $d_1 = O(1+\Lambda)$, and, by induction, $d_j = O_j((1+\Lambda)^{3j-4})$ for $j \geq 2$. Finite Taylor expansion about X then bounds the first uncanceled log-residual, and the terms discarded on squaring, at the same or smaller order. The derivative with respect to $s = 2x^2$ is comparable to Λ by (7.1). The mean value theorem therefore gives the error in (7.7). Computing d_0 and d_1 gives (7.8); its sharper error follows by including the P_1/x contribution in the residual bound. $\square$

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The Lambert normalisation (7.3) is the inverse-Gamma core of the repository's transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, section `p6:sec:gamma`): with $T = X^2$ the identity $X^2(2\Lambda - 1) = y$ reads $T(\log T - 1) = y$, which is `p6:eq:gamma-core-eq` with $R = y$, solved there by $w = W_0(R/e)$, $T = R/w$, exactly as in (7.3). The general Lambert block is `p0:thm:lambert-core` (and `p6:lem:core`), and the triangular recurrence (7.5) is a reversion around that core of the kind proved in `p0:thm:lambert-centered` and `p0:thm:perturbed-inversion`. No novelty is claimed for the method. Only the core is shared: the lower-order terms of $\log I_s$, from $x(L+1)$ on, are not those of $\log \Gamma$, so the volume's recurrence for the inverse Gamma function does not apply as it stands; this report's own inverse is (7.7).

7.2 Implicit inverses at arbitrary order

For fixed $R \geq 0$ let

$$G_R(s) = \log \mathcal{M}_s + \sum_{j=1}^R P_j (\log \sqrt{s/2}) (s/2)^{-j/2},$$

and let $N_R(y)$ be its unique sufficiently large inverse. The same mean value argument gives

$$\nu(y) = N_R(y) + O_R\left((\log N_R)^{3R+2} N_R^{-(R+1)/2}\right). \quad (7.9)$$

Using the exact-saddle approximation (4.2) instead gives implicit inverse error $O_R(s^{-R-1}/\log s)$. Its eventual invertibility follows directly by differentiating the saddle equation: $\tau'_s = 1/(\tau_s H_+)$, $H'_+ = O(\tau_s^{-2})$, and $C'_r = O_r(\tau_s^{-2r-2})$. Thus the logarithm of the truncated saddle expression has derivative $\log \tau_s + O(\tau_s^{-2}) > 0$. These are asymptotic error estimates, with unspecified constants, rather than automatic finite- n certificates.

7.3 Integer thresholds require an extra bound

The restricted-growth model gives an injection from size n to size $n+1$: append a new record $k+1$. Hence a_n is nondecreasing. Define

$$T(y) = \min\{n \geq 0 : a_n \geq e^y\}.$$

An unconditional formula $T(y) = \lceil \nu(y) \rceil$ is not justified, because the exponentially small signed contribution may matter when $\nu(y)$ is extremely close to an integer.

Here is a valid certification rule. Suppose $[\ell, u]$ is a certified interval containing $\nu(y)$. In a neighborhood containing all relevant neighboring integer indices, suppose

$$|\log a_n - \log I_n| \leq \eta, \quad (\log I_s)' \geq m > 0$$

are certified bounds. Set $\varepsilon = \eta/m$. Then, for sufficiently large y within that neighborhood,

$$\lceil \ell - \varepsilon \rceil \leq T(y) \leq \lceil u + \varepsilon \rceil. \quad (7.10)$$

Indeed the inverse points at levels $y - \eta$ and $y + \eta$ bracket the possible threshold, and each lies within ε of $\nu(y)$. Equal endpoints in (7.10) certify the integer. A proved asymptotic O estimate with an unknown constant does not by itself supply η , m , or a certified interval.

[write] **Added 2 October 2026, batch 77.** This rule has the shape of the separation condition of the same volume, `p0:thm:staircase (2)`: equal ceilings at the two ends of an interval certify the integer, here with the interval $[\ell, u]$ widened by ε to absorb the signed sector. Part (1), $N_* = \lceil \nu \rceil$, would need an exact monotone interpolation of a_n , which I_s is not. The generic arithmetic of parts (1)–(2) is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`); those statements concern an arbitrary strictly monotone function and give no formal status to (7.10).

8 Reproducibility and numerical diagnostics

The companion archive contains portable Python scripts requiring only SymPy and mpmath. They make no network calls. The forward, inverse, and oscillatory coefficient generators print exact symbolic expressions as JSON and accept an arbitrary finite requested order.

The verifier checks the Bernoulli phase formula against an independent direct integrand expansion through order four, verifies inverse composition through b_4 , reproduces the first fifteen sequence values, and checks the exact signed-moment identity to sixty relative decimal digits at $n = 0, 1, 2, 3, 4, 5, 10$. Exact coefficients for further comparisons are generated by

$$A_{n,k} = A_{n-1,k-1} + kA_{n-2,k-1}, \quad A_{0,0} = 1, \quad (8.1)$$

with inadmissible indices zero and $a_n = \sum_k A_{n,k}$. This recurrence follows directly by deciding whether the final factor contributes its 1 or its kz term.

Let $E_R(n)$ be exact $\log a_n$ minus (5.6) truncated after P_R , with $R = 0$ meaning just $\log \mathcal{M}_n$. Representative diagnostics are:

n	$E_0(n)$	$E_2(n)$	$E_4(n)$
50	$3.83866 \cdot 10^{-1}$	$-2.29622 \cdot 10^{-3}$	$3.96371 \cdot 10^{-5}$
100	$3.15767 \cdot 10^{-1}$	$-1.06134 \cdot 10^{-3}$	$1.01831 \cdot 10^{-5}$
200	$2.58405 \cdot 10^{-1}$	$-4.85957 \cdot 10^{-4}$	$2.57829 \cdot 10^{-6}$
500	$1.96765 \cdot 10^{-1}$	$-1.70911 \cdot 10^{-4}$	$4.12307 \cdot 10^{-7}$
1000	$1.59245 \cdot 10^{-1}$	$-7.68743 \cdot 10^{-5}$	$1.01915 \cdot 10^{-7}$

At $y = \log a_n$, the inverse diagnostics below compare n with the explicit approximation through the indicated b_j . This also includes the tiny distinction between the inverse of I_s and the integer sequence.

n	through b_1	through b_2	through b_4
50	$-1.10738 \cdot 10^{-1}$	$-1.98776 \cdot 10^{-3}$	$1.59680 \cdot 10^{-5}$
100	$-7.89875 \cdot 10^{-2}$	$-1.08696 \cdot 10^{-3}$	$6.37038 \cdot 10^{-6}$
200	$-5.70618 \cdot 10^{-2}$	$-6.31740 \cdot 10^{-4}$	$2.47762 \cdot 10^{-6}$
500	$-3.77495 \cdot 10^{-2}$	$-3.15337 \cdot 10^{-4}$	$6.53684 \cdot 10^{-7}$
1000	$-2.78863 \cdot 10^{-2}$	$-1.85663 \cdot 10^{-4}$	$2.27180 \cdot 10^{-7}$

The exact-saddle expansion is already more accurate at low order: at $n = 1000$, the relative error after C_2 is about $1.91 \cdot 10^{-11}$. Separate quadrature diagnostics check the oscillatory sector with the positive exponential amplitude factored out, avoiding severe underflow and avoiding division by the sine.

Numerical checks support transcription and implementation; they do not prove the remainder estimates or provide interval-certified thresholds. The proof rests on the exact entire-parameter identity, concavity, finite Taylor remainders, and Gaussian moment bounds.

[write] Added 2 October 2026, batch 77: further research questions. The delivered README, which this report's README replaces, closes with four questions that the article does not print. They are kept here, as delivered in substance: (1) how far the bivariate identity (2.6) can be made uniform when the record-count fugacity varies with n , with the maximal parameter ranges, the transitions between them, and the resulting limiting or local-limit laws for the number of records (beyond the fixed-fugacity results here); (2) whether explicit remainder constants and interval quadrature can turn the smooth inverse into an efficient certified integer-threshold algorithm, including levels exponentially close to an integer crossing; (3) the large-order growth of the positive and signed coefficient families and the truncation that minimises the error, related to the exponentially small sector without assuming a complete transseries or convergence; (4) how small the signed moment J_n can be along integer subsequences, and how its sign changes and near-cancellations are governed by the slowly varying phase θ_n ; the amplitude-scaled expansion (6.6) does not give uniform relative accuracy near oscillatory zeros. No report of the collection answers any of them at this write.

A Explicit inverse coefficients

In this appendix write $L = \Lambda$ and $c = 1/2 - \log 2$. The first two coefficients are

$$b_0 = -\frac{L+1}{L}, \quad b_1 = -\frac{L^4 + 6L^3 + (8c-5)L^2 - 2L + 1}{8L^3}.$$

For the next three, define polynomials N_2, N_3, N_4 by

$$\begin{aligned} N_2 &= -L^7 - 6L^6 - 18L^5 + (24c - 35)L^4 + (24c - 12)L^3 \\ &\quad + (-24c + 9)L^2 + 7L - 3, \\ N_3 &= 8L^9 + 15L^8 + 24L^7 + (-48c - 10)L^6 \\ &\quad + (192c^2 - 240c + 173)L^4 + (-192c + 88)L^3 \\ &\quad + (144c - 42)L^2 - 40L + 15, \\ N_4 &= 9L^{13} + 75L^{12} + 105L^{11} + (-240c + 1290)L^{10} \\ &\quad + (-960c + 2610)L^9 + (-2880c + 4418)L^8 \\ &\quad + (2880c^2 - 8400c + 3525)L^7 \\ &\quad + (5760c^2 - 11280c + 5040)L^6 \\ &\quad + (2880c^2 - 3600c + 1830)L^5 \\ &\quad + (-8640c^2 + 8160c - 4290)L^4 \\ &\quad + (6000c - 2619)L^3 + (-3600c + 870)L^2 + 945L - 315. \end{aligned}$$

Then

$$b_2 = \frac{N_2}{96L^5}, \quad b_3 = -\frac{N_3}{1536L^7}, \quad b_4 = \frac{N_4}{92160L^9}. \quad (\text{A.1})$$

In particular, including all five displayed coefficients gives

$$\nu(y) = 2X^2 + b_0X + b_1 + \frac{b_2}{X} + \frac{b_3}{X^2} + \frac{b_4}{X^3} + O\left(\frac{(1 + \log X)^{11}}{X^4}\right).$$

The general remainder is deliberately conservative; the explicit low-order polynomials have smaller degrees than its worst-case bound.

B Archive contents and replay

The archive contains the report PDF and editable TeX source, together with:

- `code/replay_coefficients.py`: exact positive-sector q_j and P_j generator;
- `code/replay_inverse.py`: exact d_j and b_j generator;
- `code/replay_oscillatory.py`: exact sine and cosine coefficient generator for the signed sector;
- `code/derive_real_laplace.py`: independent direct symbolic derivation through order four;
- `code/verify_all.py`: independent composition, exact integer, moment, saddle, and inverse diagnostics;

- `code/verify_oscillatory.py` and `code/check_sector.py`: independent symbolic and amplitude-scaled quadrature checks, with output under `receipts/`;
- a README, dependency specification, and checksum manifest.

Run the generators with `-order 4` or another nonnegative finite order. Very high orders may require substantial symbolic computation. The results are finite asymptotic expansions, not convergent numerical series guaranteed to improve monotonically at arbitrary truncation orders.

[write] **Added 2 October 2026, batch 77: shipped names.** In the collection the seven scripts listed above and `independent_checks.py` are under `code/` with their delivered names, joined there by the delivery-state tools `code/build.sh` and `code/package_release.py` (delivered at the archive root). The recorded outputs delivered under `receipts/` are shipped under `data/` with the same file names, as is `data/requirements.txt`; the delivered README is replaced by the report's README, and the delivered PDF and the checksum manifest `SHA256SUMS` are not shipped (the latter was verified, 24 of 24 entries, at placement). This source is `article.tex`, delivered as `a124380-asymptotics.tex`. The checkers and generators write only to standard output and import one another from their own directory, so they run unchanged from `code/`; `build.sh` and `package_release.py` name the delivered files and must not be run in the report directory.

References

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- [2] W. Y. C. Chen, N. J. Y. Fan, and A. F. Y. Zhao, *Partitions and Partial Matchings Avoiding Neighbor Patterns*, arXiv:1009.4535, 2010. Theorems 1.1 and 2.1. <https://arxiv.org/abs/1009.4535>.
- [3] G. Cerbai, A. Claesson, and B. E. Sagan, *Self-modified difference ascent sequences*, arXiv:2408.06959, 2024. Theorem 4.5 and Lemma 4.6 in the author-hosted version consulted 1 October 2026. <https://akc.is/papers/046-Self-modified-d-ascent-sequences.pdf>.
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